Ashwin Chembu

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PROFILE

Backend engineer building Python and Node.js APIs, storage integrations, authentication, and event-driven workflows. Focuses on secure service boundaries, durable data handoffs, and repeatable validation.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, SQL

Services / APIs: FastAPI, Flask, Node.js, Express, REST APIs, OpenAPI, MCP

Storage / Cloud: MongoDB, Redis, AWS S3, Firebase, Render

Reliability: HMAC webhooks, authentication, validation, rate limiting, pytest

EXPERIENCE

Business Technology Solutions Associate | ZS Associates — South San Francisco, CA | Jun 2024-Present

- Built **Python** and Flask endpoints connecting AWS S3 and Smartsheet APIs to automate document management and cloud data handoffs.
- Built React and FastAPI RAG assistants with AWS document ingestion and Redis services to support enterprise knowledge-search workflows.
- Developed Snowflake SQL pipelines across 327K+ records for identifier matching, specialty joins, address comparison, and quality reporting.
- Contributed to enterprise RAG evaluation and model workflows using FastAPI, Uvicorn, and pytest to validate applied-AI service behavior.

Founder and Lead Developer | Tallyrus — Jun 2023-Present

- Built Flask services with **MongoDB** persistence and S3 document storage to manage application data and support essay-feedback workflows.
- Built React/TypeScript and **Express** flows with Google OAuth, **MongoDB** persistence, secure sessions, and protected application routes.
- Developed essay-feedback workflows with **Python,** LangChain, Pinecone, and OpenAI to support retrieval-augmented grading for educators.

Data Analyst / Web Development Lead | Tech4Good at UCSC — Mar 2023-Present

- Implemented 20+ attributed Node/**Express** REST routes with **MongoDB** persistence and logging for shared chat, text entry, and polling.
- Separated state across three **MongoDB** logical databases, using change streams and persisted responses to synchronize edits and votes.
- Contributed 158 attributed commits to a collaborator-owned FigJam repository, implementing TypeScript chat, polling, and export workflows.

PROJECTS

Private Filedrop | Secure File Exchange — 2026

- Built password-protected uploads and API-key-protected **Express** interfaces with **OpenAPI** documentation for secure file-exchange workflows.
- Implemented local and S3-compatible storage, migration tooling, version history, and unguessable download links for durable file handoffs.
- Exposed **MCP** tools for file organization and retrieval, with HMAC-SHA256 webhooks and timing-safe credential checks for signed service events.
- Deployed an **Express** service on Render and documented browser, REST, **OpenAPI,** and **MCP** workflows for secure cross-session file handoffs.

Forge | Personal Fitness Intelligence — Personal project

- Built FastAPI services integrating Hevy, Oura, and Apple Health webhooks into **MongoDB** to unify training, nutrition, and recovery records.
- Implemented six-hour synchronization, on-demand refresh, and CSV ingestion to maintain fitness records across external service integrations.
- Developed an engine comparing performance with a 12-week hypertrophy plan to score workout adherence, nutrition compliance, and recovery.

Lume | AI Meeting Intelligence — Dec 2025-Present

- Built React and Node/**Express** workflows with **MongoDB** and S3 to capture, store, search, and query meeting transcripts and project files.
- Designed a RAG engine with vector embeddings, intent routing, evidence extraction, and generator/critic refinement for citation-backed answers.
- Implemented reusable Project Brain knowledge bases that retain terminology and relationships from transcripts and seed files for grounded Q&A.

EDUCATION

University of California, Davis | **B.S. Computer Science** | **GPA: 3.7** — Graduated Jun 2025

CERTIFICATIONS

AWS Certified Cloud Practitioner | Google Data Analytics Certificate

LINKS

Portfolio: https://ashchembu.com | LinkedIn: linkedin.com/in/ashwin-chembu-a5292b205 | GitHub: github.com/ashwinchembu